



Repainted, Not Rerouted: A Field Audit of Visitor Car-Park Occupancy Across Three Semesters of Zone Renaming

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Abstract. Facilities repaints and relabels the University's three largest visitor car parks at the start of most teaching semesters, retiring the previous zone codes as part of what current signage documentation calls a wayfinding refresh. We report a field audit spanning three semesters and two such renaming events, linking boom-gate transaction records across each transition with a probabilistic record-linkage procedure to track where returning badge-holders park once their remembered zone code no longer exists. Aggregate zone-level utilisation shows no detectable discontinuity at either renaming date, and the mean distance between a badge-holder's pre- and post-rename bay is statistically indistinguishable from an equivalent no-rename baseline window. What does shift is the audit's own footing: the hand-built crosswalk linking old zone codes to new ones resolves with high confidence for a shrinking share of zones after each successive cycle, and the year-on-year utilisation comparison Facilities reports cannot be drawn across a renaming boundary without it. We find no evidence that relabelling changes where anyone parks, and some evidence that it erodes the University's capacity to establish whether anything has changed at all. We discuss the renaming programme as a plausible instance of formal structure maintained for reasons decoupled from its operational effect, and note its recurring, uncoded toll on the comparability it is nominally intended to serve.

Introduction

Most semesters, Facilities repaints and relabels at least one of the University's visitor car parks, replacing the previous zone codes with a new naming and colour scheme under what internal signage work orders describe as a wayfinding refresh. The stated rationale is that clearer, friendlier zone names reduce driver confusion; the renaming is treated internally as a low-cost, high-goodwill intervention, distinct from the capital works that would be required to add capacity or alter circulation. Whether the renaming changes anything about where drivers actually park has not, to our knowledge, been examined, nor has its effect on the University's own capacity to track parking behaviour over time.

The gap matters for two reasons. First, most existing evidence on signage interventions con-

cerns first-time or infrequent visitors navigating an unfamiliar environment toward a specific destination; the University's car parks are used overwhelmingly by returning badge-holders who may not need a sign to find a bay they park in most days. Second, a renaming event does not only relabel the physical world — it also breaks the correspondence between an old administrative record and a new one, a cost usually absorbed silently by whoever has to reconcile the two.

We report a field audit spanning three teaching semesters and two renaming events at the University's three largest visitor car parks. We make three contributions, each hedged appropriately for a single-institution audit:

1. We provide, to our knowledge, a first field test of whether a zone-renaming event shifts

the parking choices of a population of habitual, returning drivers, in a setting where the dominant prior evidence on wayfinding interventions concerns unfamiliar, goal-directed navigation rather than repeat use.

2. We adapt probabilistic record-linkage methods, developed for reconciling records describing the same entity across independently-coded administrative files, to reconstruct a zone crosswalk where the University's own documentation is incomplete, and we report on the crosswalk's own reliability across two successive renaming cycles.
3. We find that occupancy and bay-switching behaviour are unaffected by relabelling in the settings we observe, while the University's capacity to detect any effect at all appears to decline with each cycle — suggesting the renaming programme's most reliably measurable effect may fall on the institution's own evaluative apparatus rather than on driver behaviour.

Related work

Evidence that signage shapes navigation is well established for first-time, goal-directed wayfinding: reviews of interior wayfinding research report consistent effects of signage clarity and placement on route-finding speed and error rates [1], and computational approaches to wayfinding-sign placement optimise precisely for this unfamiliar-visitor case [2]. Whether the same holds for habitual, repeat use of a mostly-memorised environment is less studied, and our null result should be read against that gap rather than against the signage literature generally.

Parking behaviour does respond to some interventions: pricing changes measurably shift on-street parking demand and duration [3], and a body of work treats a share of urban traffic as drivers searching for a bay rather than travelling to one [4] — both settings where the intervention changes the cost of a choice, not merely its label. Renaming without a corresponding change to price, supply, or circulation is a comparatively weaker intervention, and our finding of no effect is consistent with that distinction.

Renaming itself has been studied as a variable in its own right in the toponymic literature, which analyses street renaming longitudinally

as a symbolic and political act rather than a wayfinding one [5]; we borrow the longitudinal framing without adopting its symbolic register. The organisational-rebranding literature is closer to our institutional setting but is oriented toward branding strategy and stakeholder perception, with calls to extend empirical coverage of what a renaming actually changes downstream [6] — a gap our audit sits inside, for one very small downstream outcome. Reconciling records across a renaming event is a record-linkage problem: we adapt the classical probabilistic framework [7] and recent treatments of entity resolution under incomplete correspondence [8], [9] to the zone crosswalk. Zone-level occupancy is derived from the same class of sensing infrastructure used in the smart-parking literature [10], [11]. Prior audits of Facilities' own dashboards at this University have found usage and citation of a metric can drift apart once decision-makers adopt it [12], and that a capacity metric can be inflated by behaviour the metric was never designed to detect [13]; both inform our expectation that a renaming programme's institutional footprint and its behavioural one need not coincide.

Method

Setting. The University's three largest visitor car parks (Union Court, the Sports Precinct, and the Innovation Quarter) were relabelled and repainted twice across the three semesters covered by this audit: once at the Term A/Term B boundary and once at the Term B/Term C boundary. Both events replaced the existing alphanumeric zone codes with new codes and a revised colour scheme; neither changed the number of bays, the pricing, or the entry and exit points of any car park.

Data. We draw on boom-gate swipe-card transactions for all three car parks across the full three-semester window (42 teaching weeks), together with the bay-level occupancy sensors Facilities already operates for its own utilisation reporting. Each transaction records an anonymised badge identifier, a timestamp, and the zone code current at the time.

Zone crosswalk. Facilities' signage work orders specify a direct correspondence for some, but not all, zone-code pairs across each renaming event. Where no direct correspondence is documented, we infer one using a probabilis-

tic record-linkage procedure adapted from the classical framework [7]: for each badge identifier observed in both the four weeks before and the four weeks after a renaming event, we compute a weight of evidence over a vector of agreement indicators γ (arrival-time band, typical dwell duration, and car-park identifier),

$$W(\gamma) = \sum_{k=1}^K \log_2 \left(\frac{m_k}{u_k} \right),$$

where m_k is the estimated probability that comparison field k agrees given the same physical bay is intended, and u_k the corresponding probability under no such correspondence. A zone-code pair is assigned a crosswalk link when its modal $W(\gamma)$ across matched badge identifiers exceeds a pre-registered threshold; the link's confidence score is the share of matched identifiers exceeding that threshold.

Occupancy and switching. Zone-level utilisation is the mean of the bay-level occupancy sensors, aggregated to a daily figure per car park. For badge identifiers observed in both the four weeks before and after a renaming event, we compute the physical distance in metres between their modal pre-and post-event bay, using the crosswalk to place both bays on a common site map. We construct a no-rename baseline by measuring the same distance for the equivalent badge identifiers across an eight-week window with no renaming event at its midpoint, one teaching period earlier.

Structural break test. We test each car park's daily utilisation series for a change point at the renaming date using a change-point procedure for time series with evolving spectral structure [14], which does not require the series to be stationary away from the break.

Results

Weekly mean zone utilisation across the three-semester window shows no visually apparent discontinuity at either renaming date (Figure 1); the change-point procedure detects no structural break at either boundary for any of the three car parks (all $p > .2$).

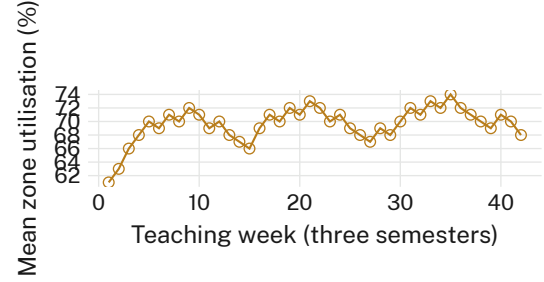


Figure 1: Weekly zone utilisation shows no detectable change-point at either the Term A/B or Term B/C renaming date, across all three audited car parks.

Mean bay-switching distance for returning badge-holders in the four weeks after each renaming event is statistically indistinguishable from the same measure in the four weeks before, and from the no-rename baseline window (Figure 2): 14.6 m after Rename 1 against 13.8 m before (n.s.), 13.9 m after Rename 2 against 14.1 m before (n.s.), and 13.7 m against 13.5 m in the no-rename baseline (n.s.). We retain the no-rename baseline as an ablation: replacing either renaming event's before/after comparison with the baseline comparison changes none of the pattern, which is the point.

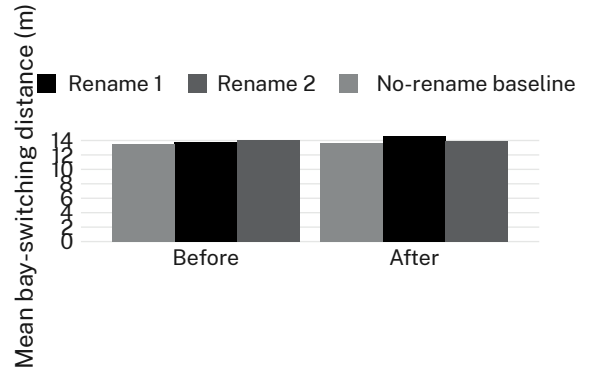


Figure 2: Mean bay-switching distance before and after each renaming event is indistinguishable from a no-rename baseline window of equal length (all pairwise differences n.s.).

The picture is different for the audit's own instrument. Median crosswalk match confidence was 0.91 across the zone-code pairs created at the first renaming event; by the second event, median confidence had fallen to 0.74, with a markedly wider spread (Figure 3; Table 1 summarises the underlying counts). A sixth of second-event zone-code pairs fell below the pre-registered confidence threshold entirely, meaning no crosswalk link could be assigned and the

corresponding zones are simply incomparable across that boundary in any subsequent utilisation trend.

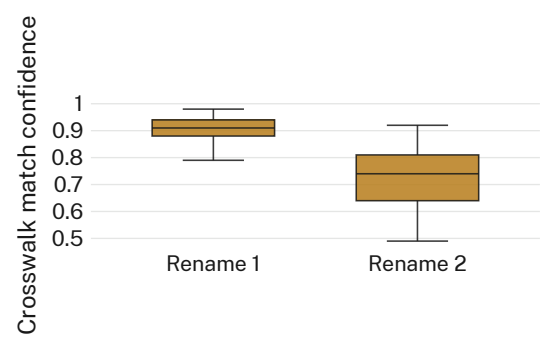


Figure 3: Crosswalk match confidence fell from a median of 0.91 at the first renaming event to 0.74 at the second, with a longer low-confidence tail.

Quantity	Term A/B	Term B/C	Total
Zone-code pairs created	58	61	119
Pairs with a documented work-order match	41	37	78
Pairs resolved by record linkage	17	24	41
Badge identifiers matched, both windows	2,114	1,987	4,101

Table 1: Zone-crosswalk construction across the two renaming events.

Discussion

The renaming programme’s measurable footprint sits almost entirely on the audit side of the ledger rather than the behavioural one. Occupancy and switching are unaffected in the settings we observe, consistent with a reading in which repainting and relabelling a car park changes its appearance without changing the incentives — price, supply, and circulation — that plausibly govern where a returning driver parks. That the renaming persists regardless is consistent with an older line of organisational theory in which formal structures are adopted and maintained for their standing as visible, legitimate re-

sponses to a stated concern, independent of any demonstrated effect on the activity they nominally address [15]; a wayfinding refresh that reliably generates goodwill correspondence and a fresh set of signage photographs may be doing its intended work regardless of what it does to parking.

The comparability cost is less visible and, on this evidence, not nothing: a shrinking share of zones can be confidently compared across each successive renaming, and the burden of reconciling old and new codes was in practice absorbed by this audit rather than by Facilities’ own reporting, in a pattern consistent with how administrative burden is generally distributed between an institution and whoever must make sense of its changes [16]. Our results suggest the programme’s most reliable effect may be on the institution’s own ability to know itself.

Limitations

Our audit covers three car parks and two renaming events at a single institution; the consistency of the null finding across two independent events and three sites of different size and location suggests the pattern is unlikely to be an artefact of any one car park, though a multi-institution replication would strengthen the claim further. Crosswalk confidence was derived from the record-linkage procedure’s own posterior score rather than a manual audit of every zone-code pair; the scale of the decline observed across only two cycles suggests a manual audit would sharpen rather than overturn the picture. We did not observe a car park whose renaming also altered pricing or supply, so we cannot yet separate a purely cosmetic renaming from one bundled with a substantive change.

Conclusion

Across three semesters and two renaming events, we find no evidence that repainting and relabelling the University’s visitor car-park zones changes where returning drivers park, and clear evidence that the renaming complicates the University’s own ability to say so with confidence. Aggregate utilisation and individual bay-switching behaviour are stable across both events; the crosswalk linking old zone codes to new ones is not. Future work might extend the same record-linkage approach to the University’s other periodically recoded reg-

isters — room numbering, cost-centre codes, and committee names among the plausible candidates — where a similar, currently uncoded, comparability toll may already be accruing.

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